1 **(a)** Hexagon **[B1]**

(b) Obtuse **[B1]**

(c) Equilateral **[B1]**

[Total Marks: 3]

2 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Minor segment (Accept segment) **[B1]**

(d) Isosceles **[B1]**

[Total Marks: 4]

3 **(a)** **(i)** $(n-2) \times 180 = 1080$ or $1080 \div 180$ **[M1]**

8 **[A1]**

(ii) Octagon **[B1]**

(b) Rhombus **[B1]**

(c) Cylinder **[B1]**

[Total Marks: 5]

4 **(a)** Kite **[B1]**

(b) Sector (or minor sector) **[B1]**

(c) Chord **[B1]**

(d) Congruent **[B1]**

[Total Marks: 4]

5 **(a)** Tangent **[B1]**

(b) Chord **[B1]**

(c) Segment **[B1]**

(d) Sector **[B1]**

(e) Reflex **[B1]**

[Total Marks: 5]

6 **(a)** **(i)** Exterior angle $= 180^\circ - 144^\circ = 36^\circ$ **[M1]**

$\frac{360}{36} = 10$ **[A1]**

(ii) Decagon **[B1]**

(b) **(i)** Sum of interior angles $= (6-2) \times 180^\circ = 720^\circ$ **[B1]**

$x + 2x + (x + 30) + (3x + 60) + 110 + 100 = 720$ OR $7x + 300 = 720$ **[M1]**

$x = 60$ **[A1]**

(ii) Right angle (since $60 + 30 = 90$) **[B1]**

(iii) Reflex angle (since $3(60) + 60 = 240$) **[B1]**

[Total Marks: 8]

7 **(a)** Frustum **[B1]**

(b) **(i)** Similar **[B1]**

(ii) Scale factor $= \dfrac{20+10}{10} = 3$ or $\dfrac{10}{30} = \dfrac{1}{3}$ **[M1]**

$r = \dfrac{12}{3} = 4$ **[A1]**

(iii) Radius difference $= 12 - 4 = 8$ **[M1]**

$\sqrt{20^2 + 8^2}$ **[M1]**

21.5 or 21.54... **[A1]**

[Total Marks: 7]

8 **(a)** Isosceles **[B1]**

(b) $138 + 180$ or $360 - (180 - 138)$ **[M1]**

318 **[A1]**

(c) **(i)** Perpendicular bisector **[B1]**

(ii) Angle $BAC = 138 - 62 = 76^\circ$ **[M1]**

Bearing $= 062 + (76 \div 2) = 100^\circ$ **[A1]**

(d) Trapezium **[B1]**

[Total Marks: 7]

9 **(a)** Angle ADE = Angle ABC because they are corresponding angles **[B1]**

Angle A is common to both triangles OR Angle AED = Angle ACB (corresponding angles) **[B1]**

(b) Length scale factor $= \dfrac{6}{10}$ or 0.6 **[M1]**

$DE = 15 \times 0.6 = 9$ **[A1]**

(c) $AC = \dfrac{7.5}{0.6} = 12.5$ **[M1]**

$EC = 12.5 - 7.5 = 5$ **[A1]**

(d) Trapezium **[B1]**

(e) Area scale factor $= 0.6^2 = 0.36$ **[M1]**

Area of triangle $ABC = \dfrac{18}{0.36} = 50$ **[M1]**

Area of $DBCE = 50 - 18 = 32$ **[A1]**

[Total Marks: 10]

10 (a) Chord **[B1]**

(b) Tangent **[B1]**

(c) Sector **[B1]**

(d) Acute **[B1]**

[Total Marks: 4]

11 (a) Kite **[B1]**

(b) Perpendicular **[B1]**

(c) Congruent **[B1]**

(d) Obtuse **[B1]**

(e) Perpendicular bisector **[B1]**

[Total Marks: 5]

12 (a) Setting up an equation connecting interior and exterior angles, e.g., $4x + x = 180$ **[M1]**

Exterior angle $= 36°$ or Interior angle $= 144°$ **[A1]**

$n = 10$ **[A1]**

Decagon **[B1]**

(b) Faces $= 12$ **[B1]**

Edges $= 30$ **[B1]**

Vertices $= 20$ **[B1]**

(c) Finding the linear scale factor $\sqrt{\dfrac{16}{9}}$ or $\dfrac{4}{3}$ **[M1]**

Finding the volume scale factor $\left(\dfrac{4}{3}\right)^3$ or $\dfrac{64}{27}$ **[M1]**

256π **[A1]**

[Total Marks: 10]